THE FORECAST HORIZON AND LONG-TERM FORECASTING

Learning Outcome

- explain considerations in the choice of an explicit forecast horizon and an analyst's choices in developing projections beyond the short-term forecast horizon

The choice of the forecast time horizon can be influenced by certain factors, including the investment strategy for which the security is being considered, the cyclicality of the industry, company-specific factors, and the analyst's employer's preferences. Most professionally managed investment strategies describe the investment time frame, or average holding period, in the stated investment objectives of the strategy; the time frame should ideally correspond with average annual turnover of the portfolio. For example, a stated investment time horizon of three to five years would imply average annual portfolio turnover between 20 percent and 33 percent (average holding period is calculated as one/portfolio turnover). The cyclicality of the industry could also influence the analyst's choice of time frame because the forecast period should be long enough to allow the business to reach an expected mid-cycle level of sales and profitability. Similar to cyclicality, various company-specific factors, including recent acquisition or restructuring activity, can influence the selection of the forecast period to allow enough time for the realization of the expected benefits from such activity to be reflected in the financial statements. In other cases, there might be no individual analyst choice in the sense that the analyst's employer has specified more or less fixed parameters. Much of the discussion so far has focused on various methods of forecasting a company's income statement, balance sheet, and cash flow for an explicit short-term forecast period. Although the underlying principles remain the same if one extends the time horizon, certain considerations and choices are available to the analyst when developing longer-term projections.

Longer-term projections often provide a better representation of the normalized earnings potential of a company than a short-term forecast, especially when certain temporary factors are present. **Normalized earnings** are the expected level of mid-cycle earnings for a company in the absence of any unusual or temporary factors that affect profitability (either positively or negatively). For example, at any given point in time, a company's profitability can be influenced by a number of temporary factors, including the stage in the business cycle, recent merger and acquisition activity, and restructuring activity. Similarly, normalized free cash flow can be defined as the expected level of mid-cycle cash flow from operations adjusted for unusual items just described less recurring capital expenditures. By extending the forecast period, an analyst is able to adjust for these unusual or temporary factors and derive an estimate of earnings that the company is likely to earn in a normal year. We will consider various alternatives for two aspects of long-term forecasting: revenue forecasts and terminal value.

As with most income statement projections, a long-term forecast begins with a revenue projection, with most of the remaining income statement items subsequently derived from the level or change in revenue. Revenue projection methods were covered earlier.

Case Study: Estimating Normalized Revenue

Exhibit 33 contains 10 years of historical revenue data and four years of estimated normalized data for Continental AG, a global automotive supplier. The accompanying bar chart in Exhibit 34 graphically depicts the data and includes a trend line based on a linear regression of the data. The numerical values for each point along the trend line can be found by using the TREND formula in Microsoft Excel. The TREND formula uses

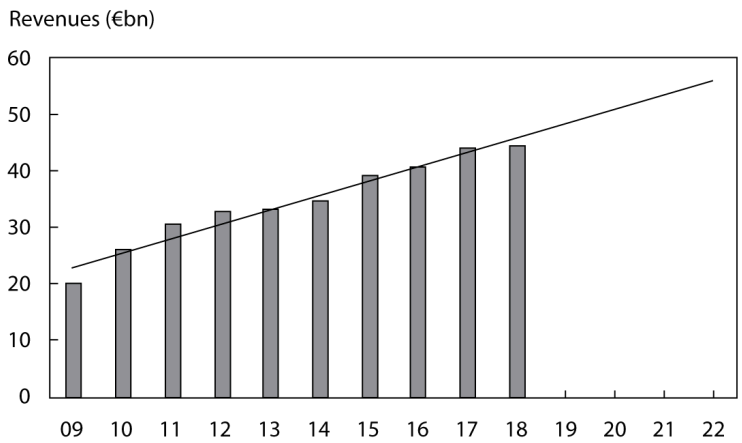
returns predicted values associated with new points in time. The worksheet for Exhibit 33 and Exhibit 34 in the downloadable Microsoft Excel workbook demonstrates the calculations used in the exhibits.

Exhibit 33:

Historical and Estimated Revenue Data for Continental AG, 2011–2024E (euro billions)

	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Revenue	30.5	32.7	33.3	34.5	39.2	40.6	44.0	44.4	44.5	37.7		
Normalized revenue	31.8	33.2	34.6	36.0	37.4	38.9	40.3	41.7	43.1	44.5	45.9	47.3
Percent above/below trend	– 4.1%	– 1.4%	– 3.7%	– 4.2%	4.8%	4.4%	9.3%	– 6.6%	3.3%	– 15.2%		

Exhibit 34: Historical and Estimated Revenue for Continental AG, 2011–2024E



▶ Expand / collapse Extended Description

The “growth relative to GDP growth” and “market growth and market share” methods discussed earlier can also be applied to developing longer-term projections. Once a revenue projection has been established, previously described methods of forecasting costs can be used to complete the income statement, balance sheet, and cash flow statement.

If an analyst is creating a valuation model such as a DCF model, estimating a terminal value is required to capture the going-concern value of the company after the explicit forecast period. Certain considerations should be kept in mind when deriving the terminal value based on long-term projections.

First, an analyst should consider whether the terminal year free cash flow projection should be normalized before that cash flow is incorporated into a long-term projection. For example, if the explicitly forecasted terminal year free cash flow is “low” (e.g., because of business cycle reasons or capital investment projects), an adjustment to normalize the amount might be warranted. Second, an analyst should consider whether and how the future long-term growth rate will differ from the historical growth rate. For example, even some mature companies might be able to accelerate their long-term growth rate through product innovation and/or market expansion (e.g., Apple), whereas other seemingly well-protected “growers” could experience an unanticipated decline in their business as a result

future will look significantly different from the recent past. Most DCF models rely on a perpetuity calculation, which assumes that the cash flows from the last year of an explicit forecast grow at a constant rate forever. Because the perpetuity can account for a relatively large portion of the overall valuation of the company, it is critical that the cash flow used is representative of a "normalized" or "mid-cycle" result. If the analyst is examining a cyclical company, using a boom year as the starting point for the perpetuity could result in a grossly overstated valuation. Similarly, using a trough year could result in a valuation that is much too low.

Another important consideration is economic disruption. The economy can occasionally experience sudden, unprecedented changes that affect a wide variety of companies, such as the 2008 global financial crisis or the COVID-19 pandemic. Even a company with a sound strategy and solid operations can be thrown far off course by a sudden economic disruption, particularly if the company has a high degree of financial leverage.

Regulation and technology are also potential drivers of inflection points, and it is important for the analyst to keep a close eye on both. Government actions can have extreme, sudden, and unpredictable impacts on some businesses. Technological advances can turn fast-growing innovators obsolete in a matter of months. Both regulation and technology affect some industries more than others. Utilities experience intense regulation but might not see a significant technological change for decades. Semiconductor manufacturers must constantly keep up with new technology but experience relatively light regulation. Pharmaceutical manufacturers are heavily exposed to both regulation and technological advances.

Finally, long-term growth is a key input in the perpetuity calculation. Some companies and industries can grow faster than the overall economy for long periods of time, causing them to account for an increasing share of overall output. Examples include some technology companies, such as Tencent, Amazon, and Google. Other companies, such as those in the print media sector, are likely to grow slower than the overall economy or even shrink over time. Using an unrealistic long-term growth rate can put the analyst's valuation far off the mark.

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EXAMPLE 10

Important Considerations When Making Assumptions

1. Turkish Airlines (THYAO.IS) operates in the highly cyclical global airline industry. Operating margins for 2011–2019 are shown in the following table and in the Example 10 worksheet in the downloadable Microsoft Excel workbook.

	2011	2012	2013	2014	2015	2016	2017	2018	2019
Operating margin	1.0%	10.8%	6.5%	5.6%	8.6%	−2.9%	9.0%	9.9%	7.9%

On the basis of only the information in the table, which of the following operating margins would *most likely* be appropriate to use in a perpetuity calculation for Turkish Airlines to arrive at a reasonable intrinsic value estimate?

A. 6.0 percent
B. 9.0 percent
C. 9.9 percent

Hide Solution

Solution:

For each of the companies in the following problems, indicate which of the choices is *least likely* to cause a change in the company's outlook.

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2. ABC Diesel (hypothetical company), a manufacturer of diesel-power trucks

A. Consumers have started switching to trucks with electric engines, threatening ABC's historic strength in diesel engine trucks.
B. ABC Diesel has formed a partnership with Electrico (hypothetical), a company involved in research and innovation in electric engines.
C. Environmental regulations have been getting tighter in most regions, and consistent with prior experience, this need to make the engines less polluting is expected to continue over the next several years.

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Solution:

C is correct. Although it is important that environmental regulations have been getting stricter, this is consistent with past experience and so does not represent a turning point.

3. Abbott Laboratories, a diversified manufacturer of health care products, including medical devices

A. Management reiterates its long-standing approach to capital deployment.
B. A competitor has demonstrated favorable efficacy data on a medical device candidate that will compete with an important Abbott product.
C. It has become more difficult for medical device manufacturers to receive regulatory approval for new products because of heightened safety concerns.

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Solution:

A is correct. Management is sticking with its historical approach to capital deployment, so this does not represent a turning point.

4. Grupo Aeroportuario del Sureste, operator of nine airports in Mexico, especially in the tourist-heavy southeast

A. Global economic disruption has caused a sharp decline in international travel.
B. Regulators will allow the construction of a new airport by a competitor in Grupo Aeroportuario del Sureste's service territory.
C. A technological advance will allow airlines to save 5 percent on fuel costs, but it is not expected to meaningfully alter passenger volumes. Similar developments in the past have benefited airlines but not airports, whose price per passenger is regulated.

Hide Solution

Solution:

C is correct. Although the technological advance is good for the airlines, it will not have a meaningful effect on passenger volumes, which will likely prevent the airports from sharing in that benefit. In contrast, both A and B could have a significant impact on the long-run earnings power of Mexican airports.

5. LinkedIn, operator of an online social network for professionals and part of Microsoft Corporation, with limited investment needs and no debt

A. Facebook, another online social network, announces a plan to enhance its offerings in the professional category.
B. Regulators announce an investigation of LinkedIn's privacy practices, which could result in significant changes to the service.

Hide Solution

Solution:

C is correct. Because LinkedIn carries no debt, it is unlikely that higher interest rates will cause a change in the company's outlook.

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OPTIONAL UNTESTED SEGMENT

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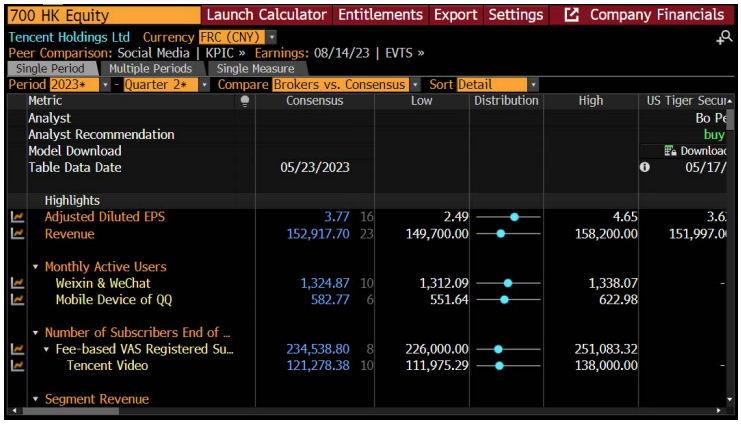
Bloomberg is one of the main market data platforms used globally by finance professionals throughout their workflows. In this optional section, we will explore a real-world application of the CFA Curriculum on Bloomberg.

{MODL} COMPANY FINANCIALS

The Bloomberg MODL function allows you to view the rows that a usual bottom-up analyst model will include for a selected company.

> "MODL <GO> provides reported actual data, broker estimates, and Bloomberg estimates consensus data for a security. You can quickly compare metrics across all key measures, including segments, products, and company-specific operating metrics to understand what's driving a company's business." - Bloomberg

To explore this function, type MODL into the Bloomberg command line and press Enter or <GO>. Type a security—such as Tencent Holdings Limited (or 700 HK Equity)—into the top left amber field, and press Enter or <GO>.



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